

CLEAVE SNP Final Project Report

Team Name: CLEAVE SNP

Team Members: Luke Hovagimian, Alexander Leong, Minh Nguyen, Heewon Park

Project Description

CLEAVE SNP is a computational biology platform that detects disease-causing SNPs and suggests CRISPR-based genetic corrections. The user can paste in a DNA sequence or upload a FASTA file, and the system compares it against a healthy reference sequence using BLAST. From there, the platform identifies known disease-related variants and displays them in an easy-to-read results page. After SNP detection, the system generates candidate guide RNAs that could be used for correction. We also included export features so the results can be downloaded in different formats. Overall, the goal of this project was to build one complete workflow that goes from sequence input to mutation detection to possible CRISPR design.

Major Software Components

Frontend (*src/*)

This folder contains the user interface of the website. It handles sequence input, page navigation, displaying SNP results, showing CRISPR candidates, and exporting reports. It is built with React and organized into pages, components, styles, and shared state.

Backend (*BLAST/server/*)

This part of the project connects the frontend to the sequence analysis pipeline. It receives requests from the frontend, processes the input, runs the appropriate analysis functions, and sends the results back to the user interface. It also handles database setup, configuration, CRISPR-related routes, and exports.

BLAST Core (*BLAST/Blast_code.py*)

This module handles the main BLAST-related sequence processing tasks. It reads and writes FASTA files, normalizes DNA sequences, builds BLAST databases, runs BLAST comparisons, and manages sequence records. This file is one of the core pieces that allows the system to compare a user's sequence against the reference database.

SNP Analysis (*BLAST/snp_analysis.py*)

This component takes the BLAST alignment results and determines whether the user sequence contains known disease-associated SNPs. It maps the variant information into a structured format and also creates the formatted SNP analysis report used for export. This is the main logic behind identifying disease-causing mutations.

CRISPR Design Logic (*BLAST/server/crispr_logic.py*)

This module generates candidate guide RNAs based on the SNP selected by the user. It uses the surrounding sequence information and alignment context to propose possible CRISPR correction targets. These results are then displayed in the frontend for the user to review.

Database and Disease Variant Tools (*BLAST/disease_database_tools.py* and database routes)

These files manage the sequence databases used in the project. They allow the user to build the healthy reference database, update it, clear it, and manage the disease SNP FASTA records. This makes the system easier to initialize and maintain.

Export Functions (*BLAST/server/export_tools.py* and results pages)

These parts handle exporting results into TXT, JSON, CSV, and FASTA-style formats. The backend writes the report content, and the frontend triggers the file download. This gives users a way to save their analysis results outside the app.

Special Instructions for Running the Code

This project does not create a hardware bit file because it is a software web application, not an FPGA-based design. Instead, the project must be set up by running both the frontend and backend.

The project requires:

- Node.js 18 or newer
- Python 3.10 or newer
- NCBI BLAST+ installed separately

To install the backend dependencies, in a terminal, run:

```
pip install -r requirements.txt
```

To start the backend, run this from the project root:

```
uvicorn BLAST.server.main:app --reload --port 8000
```

It is important to run this command from the root folder, not inside the BLAST folder, because the backend uses package-relative imports.

To install the frontend dependencies, run:

```
npm install
```

To start the frontend, in another terminal, run:

npm start

After both are running, open the website and go to the Settings page. The user must enter the full path to the BLAST+ bin folder and then build the reference database using a FASTA file. A sample FASTA file is included in the project at *BLAST/sample_fasta_test/gene.fa*.

The frontend also supports mock mode for interface testing. To run the real full pipeline, `USE MOCK` in `src/data/config.js` should be set to false.

Conclusion

CLEAVE SNP brings together web development and computational biology into one complete platform. It allows the user to input a sequence, detect disease-causing SNPs, and explore possible CRISPR correction options in one workflow. The frontend makes the system easy to use, while the backend and BLAST-based analysis perform the biological processing. Overall, this project shows how software tools can be used to make mutation analysis and therapeutic design more accessible and organized.